

REM2 Research Methods

Exercise 6-1: REIT returns and inflation

Instructions: Work in groups of two to complete the following tasks

1. Open “Lectures 4-6 Exercises Data.xlsx” in a software of your choice. Use the data in the “Lecture 6” worksheet in this exercise.
2. Generate time series charts for the return and price series. Arrange the charts in a 2 by 3 grid.
3. Plot REIT_price, Anxious_index and CPI in one time series line chart, using quarter as the horizontal axis label.
4. Check the stationarity of each variable.
5. Estimate the following linear regression model:

$$REIT_return_t = \beta_0 + \beta_1 Anxious_index_t + \beta_2 Real_GDP_t + \varepsilon$$

6. Estimate the following autoregressive model

$$REIT_return_t = \beta_0 + REIT_return_{t-1} + \beta_2 Anxious_index_t + \beta_3 Real_GDP_t + \varepsilon$$

7. Check integration among REIT_price, Anxious_index, CPI Real, and_GDP.
8. Estimate a vector autoregressive model with REIT_price, Anxious_index, CPI Real, and_GDP.
9. Estimate a vector error correction model with REIT_price, Anxious_index, CPI Real, and_GDP.